

Sprint 2 Documentation

Community Share – Student Skill Exchange Platform

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Module: CMP-N204 Software Engineering

1. Refined Project Overview

Community Share is a full-stack web application designed to support structured skill exchange among university students. The platform enables students to offer and request academic or practical skills without financial transactions, promoting mutual benefit, collaboration, and community development.

The primary problem addressed by this project is that many students require academic support, interview preparation, programming assistance, or language practice but may not have the financial means to access paid tutoring services. At the same time, many students possess valuable skills that remain underutilized. Community Share aims to connect these students in a safe and organized digital environment.

Users of the system can create a profile, list of skills they are able to offer, browse skills shared by other students, and submit skill exchange requests. The system will categorise skills (e.g., Programming, Languages, CV Review, Interview Practice, Mathematics) to improve usability and search efficiency.

The Minimum Viable Product (MVP) for Sprint 3 will include:

- User registration and login
- User profile creation
- Skill listing creation

- Browsing and filtering skill listings
- Viewing detailed skill pages
- Basic request functionality

In later sprints, the system will be extended with trust-building features such as ratings, a basic matching algorithm, and possibly in-app messaging to enhance user interaction and platform safety.

The application is implemented using Node.js and Express.js for backend logic, MySQL for persistent data storage, and PUG for dynamic templating. Docker is used to ensure a consistent development environment across all team members, and GitHub Actions will be introduced to support CI/CD workflows.

By focusing specifically on skill exchange within a university community, Community Share differentiates itself from general social media platforms by providing structured functionality, clear user roles, and a cooperative non-financial exchange model.

2. Prioritized User Stories

User stories are structured using the format:

As a [type of user], I want to [perform an action], so that [benefit].

The features have been prioritized using a MoSCoW approach (Must Have, Should Have, Could Have) to clearly define the scope of the Minimum Viable Product (MVP) for Sprint 3 and future development.

2.1 Must Have (Sprint 3 – MVP)

User Account Management

1. As a student, I want to register an account so that I can participate in skill exchanges.
2. As a user, I want to log in securely so that my personal information is protected.
3. As a user, I want to view and edit my profile so that others can understand my skills and background.

Skill Listings

4. As a user, I want to create a skill listing so that I can offer help to others.
5. As a user, I want to edit or delete my skill listing so that I can keep my information accurate.
6. As a user, I want to browse available skill listings so that I can find support.
7. As a user, I want to filter skills by category so that I can quickly find relevant help.
8. As a user, I want to view detailed information about a specific skill so that I understand what is offered before making a request.

Skill Requests

9. As a user, I want to send a skill request so that I can arrange an exchange.
10. As a user, I want to view my incoming and outgoing requests so that I can manage my exchanges.
11. As a user, I want to accept or reject a request so that I can control who I exchange skills with.

Administration (Basic Moderation)

12. As an admin, I want to view all users so that I can monitor platform activity.
13. As an admin, I want to remove inappropriate skill listings so that the platform remains safe and professional.
14. As an admin, I want to suspend a user if necessary, so that misuse of the platform can be controlled.

2.2 Should Have (Sprint 4)

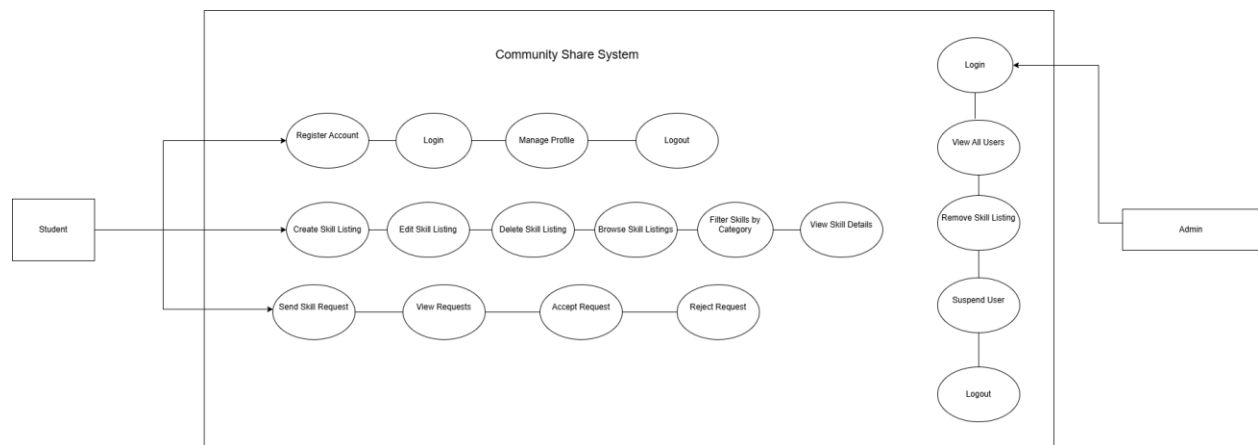
- As a user, I want to rate another user after an exchange so that trust is maintained within the community.
- As a user, I want to see a user rating before sending a request so that I feel confident engaging with them.
- As a user, I want to receive notifications when my request status changes so that I stay informed.

- As an admin, I want to view reports submitted by users so that I can investigate inappropriate behavior.

2.3 Could Have (Future Enhancements)

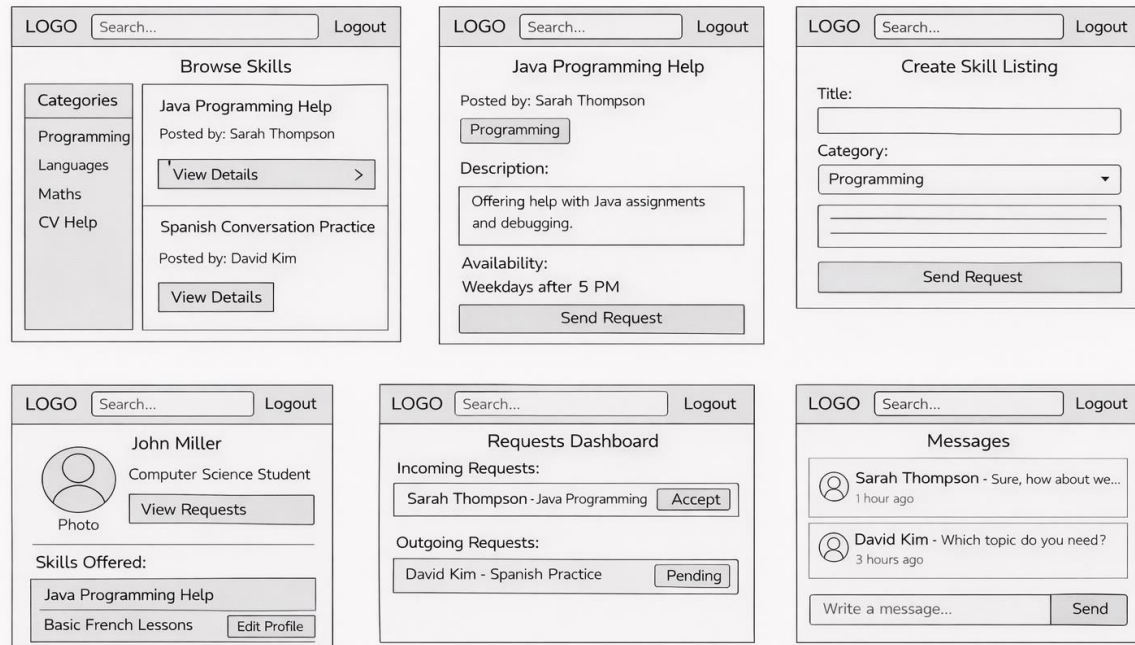
1. As a user, I want to receive skill recommendations based on my interests so that I can discover relevant exchanges.
2. As a user, I want an in-app messaging system so that communication remains within the platform.
3. As a user, I want a points or reputation system so that active contributors are recognized.

3. Use Case Diagram



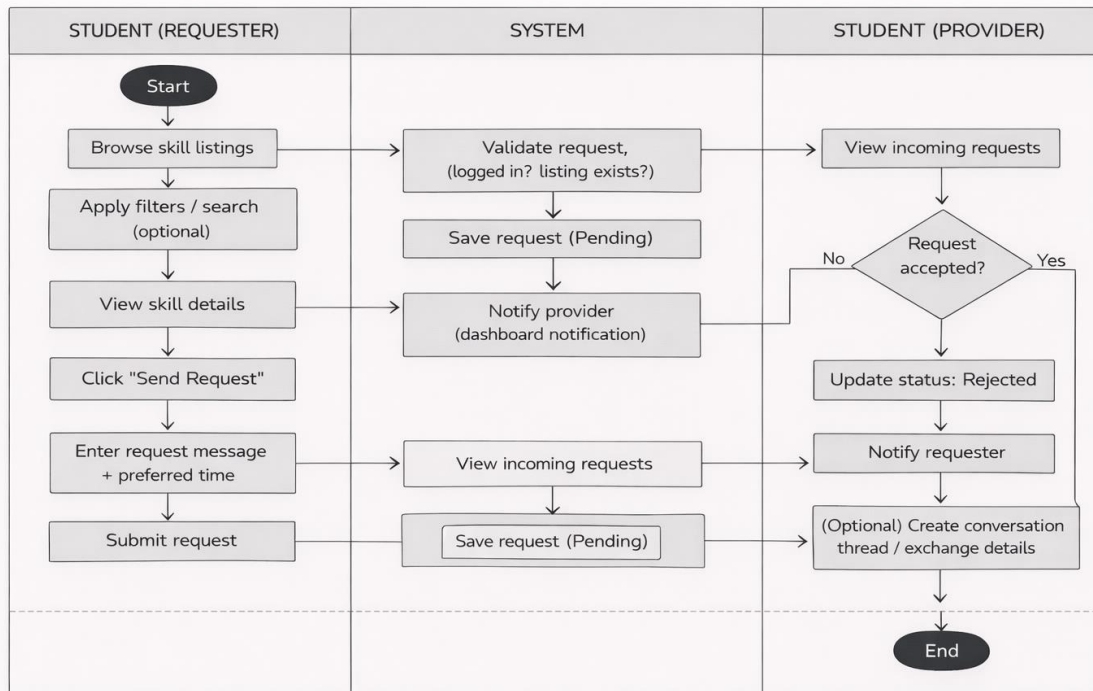
The use case diagram illustrates interactions between Student and Admin actors within the Community Share system. Students manage profiles, create listings, browse skills, and submit requests, while Admin users provide moderation functionality. The diagram reflects the functional scope of the Sprint 3 MVP.

4. Wireframes



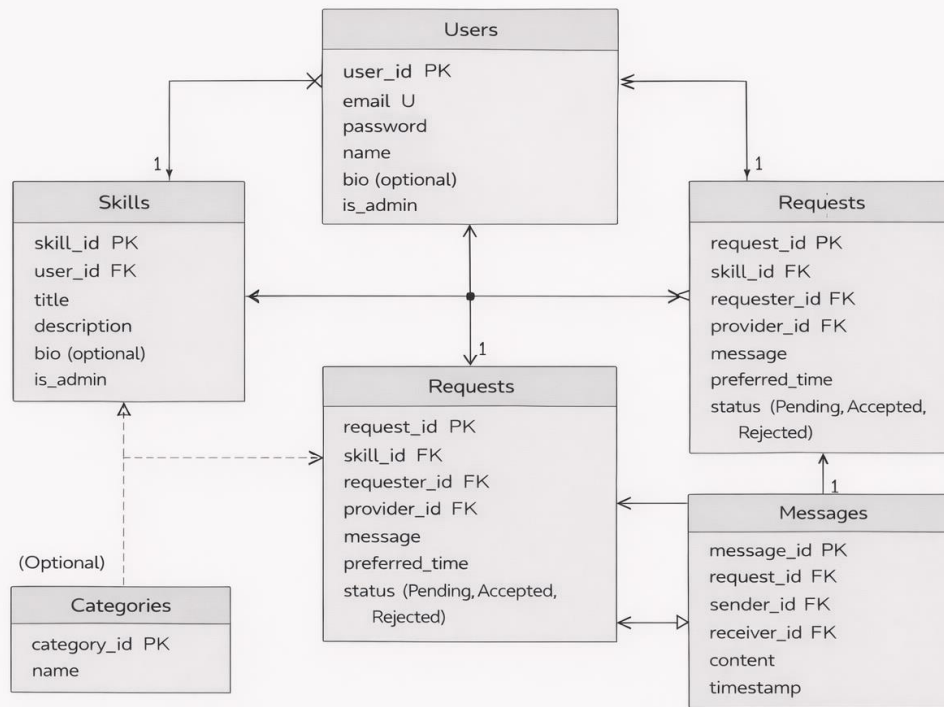
The wireframes present the structural layout of the main system interfaces, focusing on usability and navigation flow rather than visual styling. They support early design decisions before implementation.

5. Activity Diagrams / Flowcharts



The activity diagram demonstrates the workflow for submitting and managing a skill request, including validation, decision-making, and status updates.

6. Database Design (ER Diagram)



The ER diagram models the relationships between Users, Skills, and Requests. A user can create multiple skills, and each skill can have multiple requests with defined status values (Pending, Accepted, Rejected).

7. Sprint 3 Planning

- Sprint 3 Implementation Tickets

Sprint 3 focuses on the implementation phase of the Community Share system.

Based on the prioritized user stories and design artefacts developed during

Sprint 2, implementation tickets were created and organised using a Scrum-based

Kanban workflow in GitHub Projects.

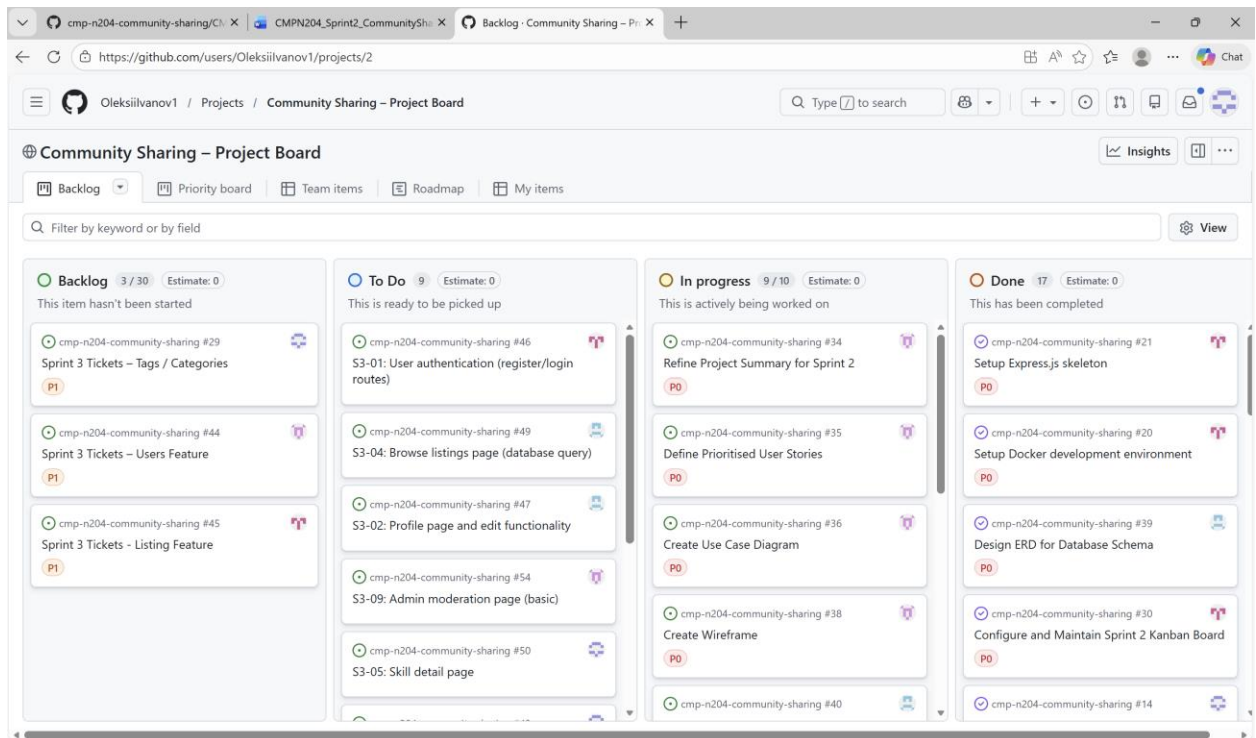


Figure 7: Sprint 3 Kanban Board showing prioritised development tickets. The board demonstrates backlog refinement, planned Sprint 3 tasks in the To Do column, active work in progress, and completed setup activities.

- Task Allocation

Following the completion of design activities in Sprint 2, the team prepared a structured implementation backlog for Sprint 3.

Each feature was converted into individual development tickets within the GitHub Project board to ensure clear task ownership, traceability, and incremental delivery.

The Sprint 3 tickets focus on core Minimum Viable Product (MVP) functionality, including user authentication, profile management, skill listing creation, browsing functionality, request handling, and basic administrative moderation.

These tasks were prioritised according to the MoSCoW framework defined earlier, with “Must Have” features placed in the To Do column to guide development order.

Breaking down features into smaller issues allowed the team to adopt an agile workflow, monitor progress transparently, and reduce the risk of overlapping responsibilities between team members.

- Screenshot of Kanban Board

The Kanban board was used to organise Sprint 3 development tasks using an agile workflow.

Tasks were prioritised and allocated to team members to support incremental implementation and progress tracking.

8. Project Links

- **GitHub Repository:**
- <https://github.com/Oleksiilvanov1/cmp-n204-community-sharing>
- **GitHub Project Board (Kanban Board):**
- <https://github.com/users/Oleksiilvanov1/projects/2>
- The repository contains the application source code, documentation, and Docker configuration used during development.
- The GitHub Project board demonstrates Sprint planning, task allocation, and collaborative workflow management following Agile Scrum practices.